

题目	一	二	三	总分
得分				
批阅人				

一、Please write a "T" or "F" in the bracket. (total 13 points)

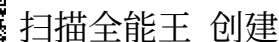
- (F) The delayed branch technique exhibits the same performance in superscalar processing as that in the RISC pipeline.
- (F) In register windows the local registers at one level are physically the same as the parameter registers at the next lower level.
- (T) Some dependencies, such as the write-read dependency, can not be solved by register renaming.
- (T) In a superscalar processor, reorder the input instruction helps to improve the speed of instruction execution.
- (F) Instruction level parallelism is a measure of the ability of the processor to deal with multiple instructions simultaneously.
- (F) Convert (ABCDE+ $\times\times/$) to infix ($(B\times C\times(D+E))/A$) .
- (F) Read-write dependency does not affect instruction level parallelism.
- (F) Computer memory is organized into a hierarchy. At the highest level (closest to the processor) are the processor caches.
- (T) There are three common places for storing the return address: Register, Start of called procedure, and Top of stack.
- (T) In immediate addressing, the operand value is present in the instruction.
- (T) A processor includes both user-visible registers and control/status registers.
- (F) The key characteristics of CISC machines: (1) a limited instruction set with a fixed format. (2) a large number of registers or the use of a compiler that optimizes register usage, and (3) an emphasis on optimizing the instruction pipeline.
- (T) The final step mentioned in the Conceptual Depiction of Superscalar Processing is referred to as committing, or retiring.



二、Fill in blanks. (1 point / blank, total 17 points)

1. Compared with the processing without pipeline, a two-stage instruction pipeline does ^{the} not necessarily cut the instruction cycle time in half because of ^{the} execute time is longer than fetch time and branch.
2. A computer has a cache and a main memory. 20 ns are required to load a word from the cache into the processor and 60 ns are required to load a word from the memory into the cache. If the hit ratio of the cache is 95%, the average word access time of the computer is $0.95 \times 20 \text{ ns} + 0.05 \times (60 \text{ ns} + 20 \text{ ns}) = 23 \text{ ns}$ ns.
3. A cache of line size of 8 Bytes contains 8K bytes. The cache can be organized into the formats of Direct Mapping, Associative Mapping, or Set Associative Mapping. If the number of set is 1, it is a Direct cache, and if the number of set is 2^{10} , it is an Associative cache.
4. The essential elements of a computer instruction are the opcodes, which specifies the operation to be performed; the source and destination operands, which specify the input and output locations for the operation; and a next instruction, which tells the CPU explicitly or implicitly where to fetch the next instruction.
5. There are 155 registers in a computer. 11 of which are left for global use and the others constitute a number of register windows. Each window is consisted of 24 register (8 parameter registers, 8 local registers, and 8 temporary registers). The register window number is 9, and the register window structure supports procedures with depth of 8 at most without window overlap.
6. RISC is abbreviation of Reduced Instruction Set Computer.
7. CISC is abbreviation of Complexed Instruction Set Computer.
8. CPU is abbreviation of Central Processing Unit.
9. ARM is abbreviation of Access Random Memory.
10. MBR is abbreviation of Memory Buffer Register.
11. PC is abbreviation of Programme Counter.
12. MIPS is abbreviation of Multiple Instructions Per Second.





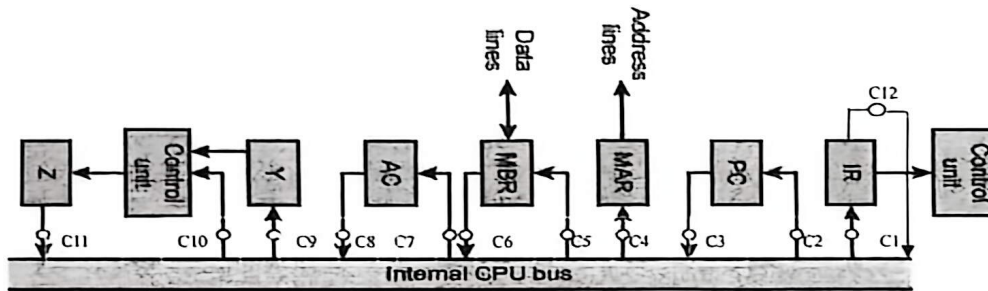


Figure 1. Internal CPU and Control Signals for a CPU

5. (12 Points) Show the micro-operations and control signals needed in the following table for the CPU in Figure 1. to fetch and execute the instruction of ADD a number to the Accumulator (the mnemonic is **ADD AC**) with 2 different addressing mode:

a. Direct address; b. Indirect address.

(Note: the cycles in Figure 1 represent gates and C with subscript is the control signals to the corresponding gates.)

Solution:

Table		
Instruction sub-cycles	Micro-operations	Active control signals
Fetch: (4 points)	$t_1: MAR \leftarrow PC$ $t_2: MBR \leftarrow Memory$ $PC \leftarrow PC + 1$ $t_3: IR \leftarrow (MBR)$	C_3, C_4 C_R C_6, C_1
Case 1 Execute (direct): (4 points)	$t_1: Y \leftarrow (IR(address))$ $t_2: Z \leftarrow (AC) + (Y)$ $t_3: AC \leftarrow (Z)$	C_{12}, C_9 C_{10}, C_8, C_{ALU} C_{11}, C_7
Case2	Indirect addressing: (4 points)	$t_1: MAR \leftarrow (IR(address))$ $t_2: MBR \leftarrow Memory$ $t_3: Y \leftarrow (MBR)$ $t_4: Z \leftarrow (AC) + (Y)$ C_{12}, C_4 C_R C_6, C_9 C_{10}, C_8, C_{ALU}
	Execute: (Note: Do Not Write)	$t_5: AC \leftarrow (Z)$ C_{11}, C_7 (Do Not Write)

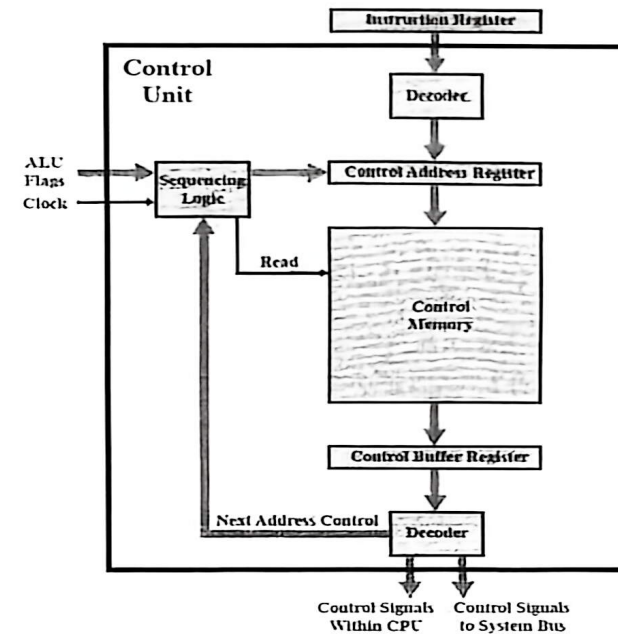


Figure 2. functioning of micro-programmed control unit

6. (10 Points) Figure 2. presents the diagram of functioning of micro-programmed control unit. Please try to answer the following questions:

a.(1 point) What is the purpose of the control memory?

Answers: The control memory contains of ^{micro}instructions that define a control

b. (3 points) How to get the next micro-instruction address?

When a clock comes, according to ALU flags & Sequencing Logic

next micro-instruction address from Instruction Register and

c. (6 points)What is the progress of the control unit to function?

When the micro-instruction address is fetched, sequencing

logic Unit can send a read signal, it can read control Memory to get the ~~in~~ micro instruction to control buffer

Register, and decoder will translate the instruction



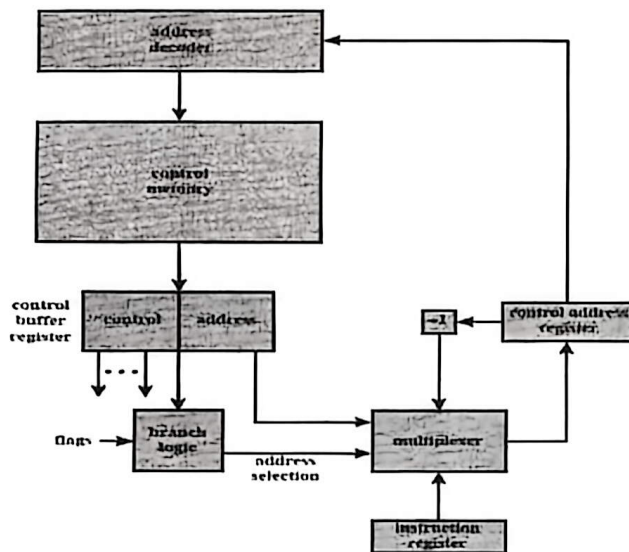


Figure 3. micro-programmed control unit

7. (4 points) Figure 3. is a control unit implementation. Assume that the control memory is 32 bits wide. The control portion of the microinstruction format is divided into two fields. A micro-operation field of 16 bits specifies the operation to be performed. An address selection field specifies a condition, based on the flags, which will cause a microinstruction branch. There are sixteen flags.

a. (1 point) How many bits are in the address selection field?

~~16~~ 4

b. (1 point) How many bits are in the address field?

~~$2^{12} \times 2 = 2^{17}$ bits~~ ~~$2^{12} \times 2 = 2^{17}$ bits~~ 12

c. (2 points) What is the size of the control memory?

$2^{12} \times 2 = 2^{17}$ bits

8. (6 points) The pipeline of a CPU has six-stage: FI (Fetch instruction), DI (Decode instruction), CO (Calculate operand's addresses), FO (Fetch operands), EI (Execute instruction), WO (Write operand). Assume there are 10 instructions. The 2nd is a conditional branch to 7th. Draw the two timing diagrams of pipeline:—

a. (4 points) If the CPU does not use Branch History Table strategy. The prediction decision is made to take the branch, and decision is correct.

b. (2 points) If the CPU uses Branch History Table strategy. The prediction decision is made to take the branch, and decision is correct.

a.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	FI	DI	CO	FO	EI	WO								
2		FI	DI	CO	FO	EI	WO							
3			FI	DI	CO	FO								
4				FI	DI	CO								
5					FI	DI								
6						FI								
7							FI	DI	CO	FO	EI	WO		
8								FI	DI	CO	FO	EI	WO	
9									FI	DI	CO	FO	EI	WO
10										FI	DI	CO	FO	EI

b.

	1	2	3	4	5	6	7	8	9	10	11			
1	FI	DI	CO	FO	EI	WO								
2		FI	DI	CO	FO	EI	WO							
3			FI	DI	CO	FO	EI	WO						
4				FI	DI	CO	FO	EI	WO					
5					FI	DI	CO	FO	EI	WO				
6						FI	DI	CO	FO	EI	WO			
7							FI	DI	CO	FO	EI	WO		
8								FI	DI	CO	FO	EI	WO	
9									FI	DI	CO	FO	EI	WO
10										FI	DI	CO	FO	EI

